

Program: Diploma in Civil and Environmental Engineering	
Course Code: 6351A	Course Title: Habitat Engineering
Semester: 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To introduce basic principles of town planning and the concept of vernacular architecture
- To impart knowledge about the effects of different climatic elements which have to be considered before fixing the orientation of a building
- To identify the principles of landscape architecture and to enable the students to prepare a landscape layout for a site
- To outline the principles of interior design and to enable the students to design an interior space

Course Prerequisites:

Topic	Course code	Coursename	Semester
Building Drawing	3358	Civil Engineering Drafting Lab	3

Course Outcomes:

COn	Description	Duration (Hours)	CognitiveLevel
CO1	Identify the principles of town planning and describe the concept of vernacular architecture	14	Understanding
CO2	Explain the concepts of climate responsive architecture, lighting, ventilation and acoustics	15	Understanding
CO3	Illustrate the landscape layout for a given plot using the principles of landscape architecture	15	Applying
CO4	Illustrate the 3D visualization of an interior space using the principles of interior design	14	Applying
	Series Tests	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3				3		3
CO2	3				3		3
CO3	3				3		3
CO4	3				3		3

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	CognitiveLevel
CO1	Identify the principles of town planning and describe the concept of vernacular architecture		
M1.01	Identify the types of settlement patterns and dwelling typologies	3	Understanding
M1.02	Identify the principles of town planning and the methods for slum clearance	4	Understanding
M1.03	Discuss the significance of vernacular architecture and identify the characteristics of the vernacular architecture of Kerala	3	Understanding
M1.04	Explain the elements and principles of architectural design	4	Understanding
Contents: Habitat, Settlement Pattern, Dwelling/Housing Typology Town Planning- Growth of towns - Principles of town planning, Zoning, Bye-laws, Master Plan Housing-Importance of housing-Slums-Types-Causes-Characteristics-Effects- Agencies and Methods of Slum clearance, Mass Housing. Vernacular architecture-Indigenous construction materials, Vernacular architecture of Kerala, Primary elements of Architectural design - Point, Line, Plane, Volume. Form - Definition, Regular and irregular forms. Principles of composition- Balance, Rhythm, Emphasis, Proportion and scale, Movement, Contrast, Unity.Space- Exterior and Interior space			
CO2	Explain the concepts of climate responsive architecture, lighting, ventilation and acoustics		

M2.01	Explain the basics of climatology and the elements of climate	4	Understanding
M2.02	Discuss the use of sun path diagrams and wind rose diagrams to fix the orientation of a building	3	Understanding
M2.03	Explain the indices of thermal comfort	4	Understanding
M2.04	Describe the concepts of day lighting, ventilation and acoustics	4	Understanding
	SeriesTest-I	1	
Contents: Climatology- Climate and Weather - Koppen classification – Macro Climate, Site Climate/ Micro Climate: - Local factors - temperature and humidity - Air movement - Vegetation - Urban Heat Island Elements of climate - temperature, humidity, precipitation, radiation, wind- Climate zones in India - Tropical Climate - Climatic design recommendations for tropical climate Sun path diagram/Solarchart - Global wind pattern - Wind movement - Wind rosedialogram. Human comfort in buildings -Thermal comfort - Indices of thermal comfort -Reduction of solar heat gain-Sol-air temperature Lighting of interior and exterior - measurement of light - Glare - Day lighting - Sky component - SP 41 Natural ventilation - Artificial ventilation- Functions of Ventilation - Factors affecting indoor airflow - airflow around buildings - Openings-Cross ventilation - Passive cooling Acoustics-sources-Soundandnoise-Noisecontrol.			
CO3	Illustrate the landscape layout for a given plot using the principles of landscape architecture		
M3.01	Define landscape architecture and identify the phytogeographical regions of India	4	Understanding
M3.02	Explain the principles and elements of landscape design	3	Understanding
M3.03	Explain the elements and features of Mughal, French, Japanese and English gardens	4	Understanding
M3.04	Sketch landscape layout for a given site	4	Applying

Contents:

Landscape Architecture: Phytogeographical regions of India

Principles - Unity, Repetition, Order, Proportion, Elements - Line, Form, Texture, Colour
Visual weight

Components of Landscape design - Softscape - Classification of Plants, Decorative vegetation, Species typical to Kerala, Lawn, Turf -Hardscape - Ferrocement, Paving blocks, Stone items, Artificially moulded elements, walkways, patios, driveways, walls, Site furniture etc. - Water as an element

Formal and Informal Gardens - Elements and features of Gardens - Mughal Gardens - English Gardens - French Gardens - Japanese Gardens (Sketch layout). Traditional Home gardens of Kerala

Bubble diagram and concept sketches

CO4	Illustrate the 3Dvisualization of a space using the principles of interior design		
M4.01	Explain the elements and principles of Interior Design	3	Understanding
M4.02	Discuss the techniques to modify interiors	4	Understanding
M4.03	Familiarize with the anthropometric data sheets for interior design	3	Understanding
M4.04	Illustrate 3Dvisualization of an interior space	4	Applying
	SeriesTest-II	1	

Contents:

Interior Space - Positive and Negative space - Circulation space - Physical quantities and Attached qualities (Tactile, Thermal, Auditory and Olfactory)

Elements of Interior design - Point, Line, Plane, Form, Shape, Texture, Colour (Colour wheel, Munsell Scale) and Light

Principles of Interior Design - Proportion, Scale, Balance, Harmony, Unity, Variety, Rhythm, Emphasis

Modification of interiors – Space modulation to modify interiors – Structural and Non-structural modulation - Artificial Lighting - Cool and Warm lighting

Classic and modern decorative materials

Furniture-Classification based on styles, materials and function – Modular furniture

Finishes - for walls, floors and ceilings - criteria for selection

Fabric-curtains,blinds,carpets,rugs

Ergonomics and Anthropometry - Activity relationships - Planning and Three Dimensional Visualization of interiors

Text/Reference:

T/R	BookTitle/Author
T1	FrancisDKChing-ArchitectureForm,Space,Order;JohnWiley&Sons,Inc
R1	Koenigsbergeret-al:Manual of tropical housing and building; Orient Longman
R2	MartinEvans:Housing,ClimateandComfort;TheArchitecturalPress, London
R3	Town Planning; Rangwala, Charotar Publication
R4	Givoni-Man,Climate and Architecture, Applied Science Publishers
R5	TimeSaverStandardsforLandscapeDesign,CharlesW.HarrisandNicholas T Dines
R6	MichaelLaurie-AnIntroductiontolandscapearchitecture,AmericanElsevier Publishing Company
R7	TimeSaverStandardsforInteriorDesignandSpace,JosephDeChiara,Julius Panero, and Martin Zelnik, McGraw-Hill, Inc

OnlineResources:

Sl.No	WebsiteLink
1	https://ncert.nic.in/ncerts/l/legyl10.pdf
2	https://www.yourhome.gov.au/passive-design
3	https://en.wikipedia.org/